

PRAKTIKUM SISTEM OPERASI

MODUL 10

SHELL



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

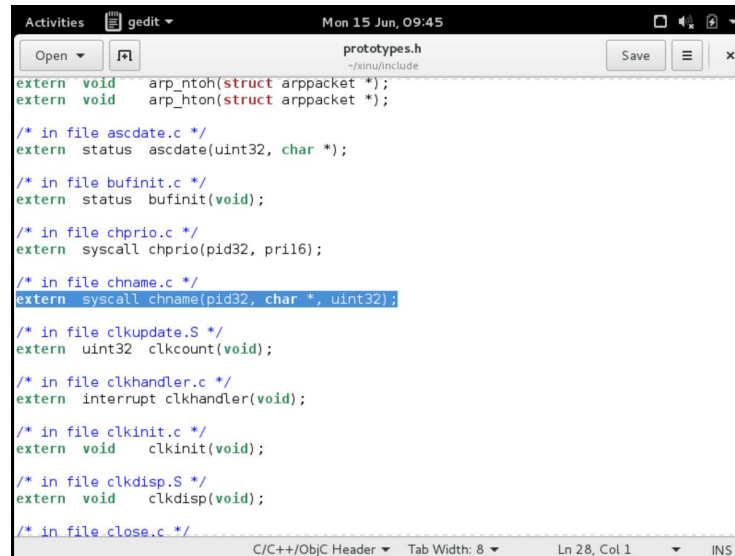
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2026

JURNAL/TP

1. Soal 1

Modifikasi syscall chname



```
Activities gedit Mon 15 Jun, 09:45
prototypes.h
~/xinu/include

extern void arp_ntoh(struct arppacket *);
extern void arp_hton(struct arppacket *);

/* in file ascdatetime.c */
extern status ascdatetime(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, prl16);

/* in file chname.c */
extern syscall chname(pid32, char *, uint32);

/* in file clkupdate.S */
extern uint32 clkcount(void);

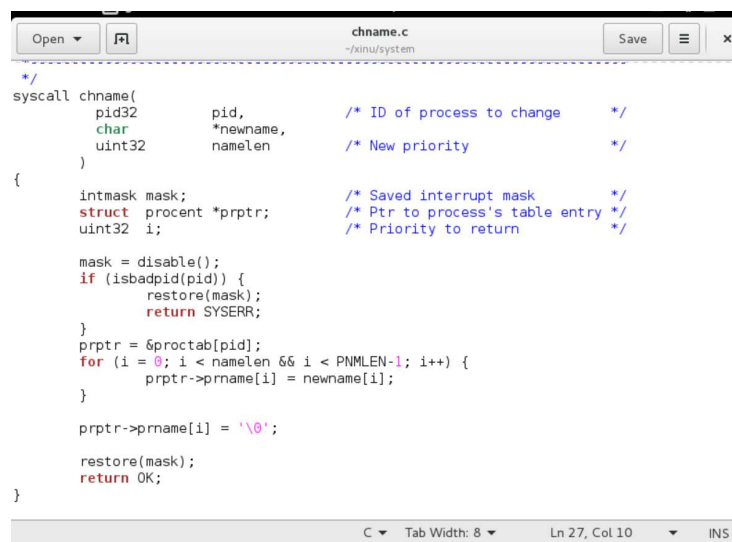
/* in file clkhandler.c */
extern interrupt clkhandler(void);

/* in file clkinit.c */
extern void clkinit(void);

/* in file clkdisp.S */
extern void clkdisp(void);

/* in file close.c */
```

Source code chname.c



```
chname.c
~/xinu/system

/*
syscall chname(
    pid32 pid,          /* ID of process to change */
    char *newname,      /* New priority */
    uint32 namelen      /* New priority */
)
{
    intmask mask;
    struct procent *prptr; /* Saved interrupt mask */
    uint32 i;              /* Ptr to process's table entry */
                          /* Priority to return */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }
    prptr = &proctab[pid];
    for (i = 0; i < namelen && i < PNMLEN-1; i++) {
        prptr->prname[i] = newname[i];
    }

    prptr->prname[i] = '\0';
    restore(mask);
    return OK;
}
```

Fungsi `chname()` digunakan untuk mengubah nama proses berdasarkan PID tertentu. Nama baru disalin ke field `prname` pada tabel proses menggunakan perulangan hingga seluruh karakter nama tersalin.

2. Soal 2

Pembuatan command baru namecmd

```
xsh_namecmd.c
~/xinu/shell

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/* NAMA : Zhafir Zaidan Avail */
/* NIM : 23111040598*/

#include <xinu.h>

shellcmd xsh_namecmd(int nargs, char *args[])
{
    printf("my new command\n");

    /* pid 2 = netin */
    chname(2, "helloww", 7);

    return 0;
}
```

Penambahan pada shprototypes.h

```
shprototypes.h
~/xinu/include

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/* in file xsh_ls.c */
extern shellcmd xsh_ls (int32, char *[]);

/* in file xsh_led.c */
extern shellcmd xsh_led (int32, char *[]);

/* in file xsh_memdump.c */
extern shellcmd xsh_memdump (int32, char *[]);

/* in file xsh_memstat.c */
extern shellcmd xsh_memstat (int32, char *[]);

/* in file xsh_namecmd.c */
extern shellcmd xsh_namecmd (int32, char *[]);

/* in file xsh_netinfo.c */
extern shellcmd xsh_netinfo (int32, char *[]);

/* in file xsh_ns.c */
extern shellcmd xsh_ns (int32, char *[]);

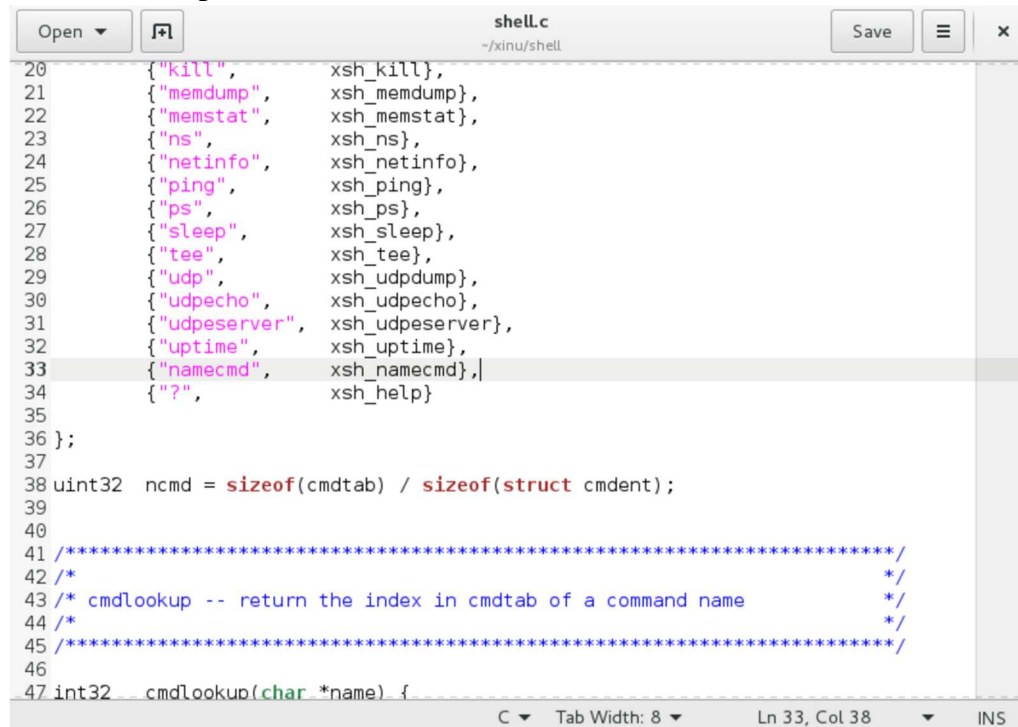
/* in file xsh_nvram.c */
extern shellcmd xsh_nvram (int32, char *[]);

/* in file xsh_ping.c */
extern shellcmd xsh_ping (int32, char *[]);

/* in file xsh_ps.c */
extern shellcmd xsh_ps (int32, char *[]);

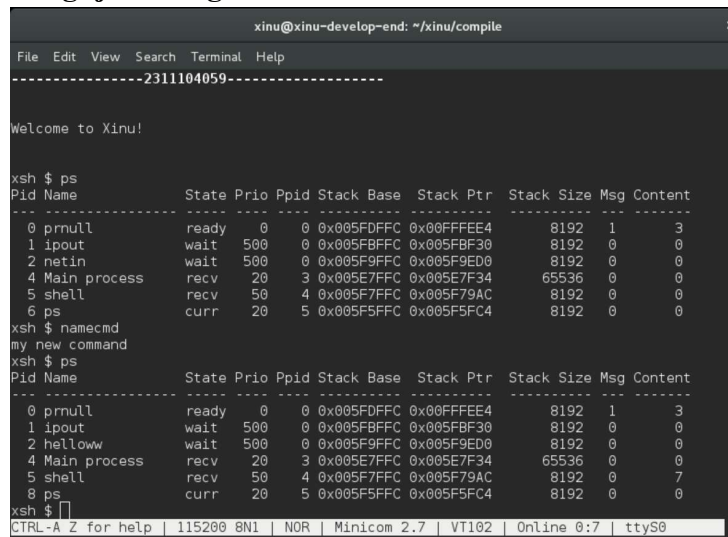
C/C++/ObjC Header Tab Width: 8 Ln 49, Col 1 INS
```

Penambahan pada shell.c



```
20 { "kill", xsh_kill},
21 { "memdump", xsh_memdump},
22 { "memstat", xsh_memstat},
23 { "ns", xsh_ns},
24 { "netinfo", xsh_netinfo},
25 { "ping", xsh_ping},
26 { "ps", xsh_ps},
27 { "sleep", xsh_sleep},
28 { "tee", xsh_tee},
29 { "udp", xsh_udpdump},
30 { "udpecho", xsh_udpecho},
31 { "udpserver", xsh_udpserver},
32 { "uptime", xsh_uptime},
33 { "namecmd", xsh_namecmd},
34 { "?", xsh_help}
35
36 };
37
38 uint32 ncmd = sizeof(cmdtab) / sizeof(struct cmdent);
39
40
41 /*
42 /*
43 /* cmdlookup -- return the index in cmdtab of a command name
44 /*
45 /*
46
47 int32 cmdlookup(char *name) {
```

3. Pengujian Program



```
xinu@xinu-develop-end: ~/xinu/compile
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base  Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0  0x005FDFFC 0x00FFFE4   8192    1    3
1 ipout       wait   500  0  0x005FBFFC 0x005FBF30  8192    0    0
2 netin       wait   500  0  0x005F9FFC 0x005F9ED0  8192    0    0
4 Main process recv   20   3  0x005E7FFC 0x005E7F34  65536   0    0
5 shell       recv   50   4  0x005F7FFC 0x005F79AC  8192    0    0
6 ps         curr   20   5  0x005F5FFC 0x005F5FC4  8192    0    0

xsh $ namecmd
my new command
xsh $ ps
Pid Name      State Prio Ppid Stack Base  Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0  0x005FDFFC 0x00FFFE4   8192    1    3
1 ipout       wait   500  0  0x005FBFFC 0x005FBF30  8192    0    0
2 helloww     wait   500  0  0x005F9FFC 0x005F9ED0  8192    0    0
4 Main process recv   20   3  0x005E7FFC 0x005E7F34  65536   0    0
5 shell       recv   50   4  0x005F7FFC 0x005F79AC  8192    0    7
8 ps         curr   20   5  0x005F5FFC 0x005F5FC4  8192    0    0

xsh $
```

Berdasarkan praktikum yang telah dilakukan, syscall chname berhasil dimodifikasi sehingga dapat digunakan untuk mengubah nama suatu proses berdasarkan PID yang diberikan. Selain itu, command baru bernama namecmd berhasil ditambahkan ke shell Xinu dan dapat digunakan untuk memanggil syscall chname. Hasil pengujian menunjukkan bahwa nama proses berhasil berubah dari netin menjadi helloww, sehingga tujuan praktikum Modul 10 dapat tercapai dengan baik.